



Evaluating Denoising Approaches for RGB-Infrared Images: Systematic Review and Comparative Analysis of Traditional Methods and Performance Metrics

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Abstract

Denoising enhances image quality by separating noise from observed signals, eliminating extraneous information while preserving essential features and image integrity. However, existing surveys on conventional denoising techniques often focus solely on processing-domain taxonomies, thereby neglecting evolutionary relationships, overlooking recent advances, and lacking multi-modal exploration. Consequently, modern machine learning pipelines have not fully exploited classical techniques. To advance multi-modal denoising and inspire new learning-based algorithms, this paper presents a comprehensive review of traditional denoising methods, quantitatively assesses their cross-modal transferability, and explores their integration into learning frameworks. Specifically, (1) this work proposes a novel taxonomy for conventional denoising techniques, including domain-based and signal-decomposition-based approaches, provides a systematic analysis of their evolutionary relationships, and investigates recent advances. (2) The study evaluates multi-modal denoising performance by applying baseline methods to infrared images and conducting a comparative analysis. (3) This paper surveys the latest research on traditional approaches, retraces their co-evolution with machine learning, and specifically explores the potential for fusing these techniques within learning-based denoising algorithms. In general, this review serves as a valuable reference for researchers in RGB-infrared denoising, image restoration, and related fields. The advancements in these areas significantly impact various domains, including defect detection in industrial production, worker protection safety recognition, and object tracking in smart transportation.

Keywords Image Quality · Image Restoration · Infrared Denoising · multi-modal Transform · Learning-based

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1 Introduction

Image noise refers to unpredictable fluctuations in visual characteristics, such as brightness and color. Noise often originates from interference within the equipment and transmission channels during signal generation and transmission [1]. Additional noise generated during the image formation process significantly impacts the quality of both infrared and visual-light digital images. Noise in general digital images caused by various factors, including external environmental conditions (e.g., lighting) and the digital equipment itself, which introduces noise during signal collection and transmission due to sensor materials, electronic components, and circuit structure. Commonly used image sensors, such as Charge-Coupled Devices (CCD) and Complementary Metal

Oxide Semiconductors (CMOS), introduce several noise types. These include thermal noise (originating from resistive and tube components), photon noise, dark current noise, and non-uniform noise arising from undesired light responses. In addition, imperfections in the transmission medium and recording equipment introduce various noise forms during signal transmission, further degrading digital image quality.

Infrared imaging noise characteristics differ from those of visible-light images, a distinction rooted in their distinct photography principles. The primary source of external environmental noise in infrared images is thermal radiation from the environment, atmosphere, and other background elements. Compared to visible-light images, equipment-related factors, such as detector components and the photoelectric conversion circuit, more significantly affect infrared image quality during acquisition and transmission. Moreover, the design of internal electrical elements for capturing thermal and RGB images differs due to their distinct operational principles. Consequently, noise generation from electrical elements during infrared image acquisition and transmission is more intricate and diverse than in visible-light images. Mixed random noise contaminates an image, resulting in numerous noisy points that reduce clarity and obscure details. This significantly obstructs the extraction of valuable information from the image. Noise reduces signal data effectiveness and profoundly impacts not only direct image information extraction but also subsequent processes such as image enhancement, target detection, and tracking.

Both digital and infrared images often contain considerable noise that affects subsequent image processing [2]. Consequently, denoising procedures are necessary for both image types. As a foundational step in image processing, denoising effectively removes unwanted noise while preserving important features and maintaining the integrity of the original image data. This process is critical for vision tasks such as target detection and semantic segmentation, as it significantly reduces unwanted artifacts, thus improving visual perception and facilitating relevant information extraction. Furthermore, image denoising contributes to refining the original signal and eliminating redundancy, thereby facilitating successful post-processing operations. Addressing image noise is a key technical challenge in image processing and has become a focus of research interest [3].

Despite significant recent research on image denoising, the field lacks comprehensive and up-to-date reviews. Existing reviews have certain limitations. Firstly, many reviews limit their scope to specific aspects of denoising methods or cover only a selection of representative algorithms, lacking a comprehensive overview [4–6]. For instance, Jiang et al. [7] focused solely on deep learning denoising techniques. Secondly, the summaries of the methods in existing reviews are often incomplete, overlooking important details

[8]. They typically provide brief introductions to concepts and formulas without adequately covering advantages, drawbacks, and experimental results [9]. Thirdly, existing reviews do not clearly explain the relationships between different methods. Existing reviews are almost solely focused on comparing ideas or performance among methods without analyzing their evolution from a historical perspective [10]. Moreover, existing reviews do not sufficiently explore recent advances in representative methods and their relationships with learning-based frameworks [11]. Lastly, most reviews primarily concentrate on RGB denoising and tend to overlook advancements in multi-modal image processing. Although some surveys consider thermal imaging, the content remains general [12].

This paper provides a comprehensive review of classical denoising techniques, covering around 200 studies, aiming to promote progress in multi-modal imaging processing and stimulate the development of learning-based algorithms. Key contributions include:

- A novel taxonomy, including domain-based and signal decomposition-based, for summarizing classical methods. Underlying concepts and theories for each category are provided, and representative methods are discussed along with analysis of their advantages and disadvantages based on performance.
- A thorough exposition of major classical denoising algorithms, tracing their evolutionary connections along a historical timeline. Specifically, latest refinements of representative methods are examined, and their interface with contemporary learning-based frameworks is clarified.
- An evaluation of multi-modal denoising by adapting baseline algorithms to infrared imagery, coupled with a comparative analysis for transferability discussion.

Section 2 examines domain-based denoising approaches, whereas Section 3 investigates methods grounded in a signal decomposition perspective. Section 4 evaluates the transferability of classical denoising techniques in thermal imagery. Section 5 reviews the latest advances in the integration of traditional techniques and deep learning frameworks, summarizes fusion strategies, and highlights promising research avenues. Finally, Section 6 summarizes the paper and discusses key findings and broader implications.

2 Domain-Based Methods

Denoising processing uses two primary approaches: classical and neural network-based algorithms [13]. Classical algorithms exploit the characteristics of image pixels and are typically categorized from two distinct perspectives. Firstly,

methods are classified based on the domain, specifically into spatial and transform domains. Secondly, they can be categorized based on signal decomposition techniques, including filtering, sparse representation, clustering, and low rank methods, depending on the sparsity level of the decomposition results. This section details domain-based classification methods, while methods based on signal decomposition are discussed in Section 3. Figure 1 shows the overarching classification framework of these classical methods according to two different criteria.

2.1 Spatial Domain

Spatial domain techniques take advantage of inherent correlations in most images [14]. Spatial filters fall into two categories: local and non-local filters. Local filters use coefficients restricted by spatial distance within a neighborhood of candidate pixels. In contrast, non-local filters are not restricted by spatial distance, but instead rely on the similarity between pixels. Historically, local spatial filters emerged as a conventional strategy, initially categorized as linear and non-linear types. Filtering technology progressed, introducing filters with combined linearity characteristics [15]. Over time, local spatial filters advanced beyond linearity, integrating non-parametric and semi-parametric filters and expanding the repertoire of techniques.

Tables 1 and 2 summarize local linear and non-linear filters, respectively, and Table 3 provides an overview of some key local spatial filters. Subsequently, non-local filters were proposed, which exploit image self-similarity, shifting the research focus from local to non-local filters in image denoising [16]. The original non-local mean (NLM) algorithm obtained a denoised patch by calculating a weighted average of image patches [17]. Since then, the NLM algorithm has undergone significant improvements [18]. Efforts

have focused on accelerating the NLM algorithm or introducing enhancements to improve quantitative and qualitative denoising performance. Despite the potential advantages of non-local filters over local filters, they can still produce artifacts such as excessive smoothing when applied to highly noisy images. Consequently, further exploration of transform domain methods is still needed [19].

2.1.1 Local Filter

Earlier studies explored various local filtering algorithms for denoising. Initially, local spatial filters were classified from the perspective of linearity, including linear and non-linear filters [20]. As filtering technology advanced, the focus shifted to parametricity. In particular, each specific filter also underwent its own longitudinal development process, as shown in Fig. 2. However, when faced with high noise levels, these methods may not yield satisfactory restoration results due to corruption of the correlations among neighboring pixels [21]. Consequently, the development of non-local filters ensued to provide better compensation; these are discussed later in this subsection.

As prototypical members of the early linear filter family, the mean filter [22], the Gaussian filter [23], and the Wiener filter [24] have undergone multi-decade trajectories of theoretical refinement and practical deployment. However, as the complexity of the noise models increases, the limitations of linear filters have gradually become evident, despite their ability to partially remove noise [25]. Firstly, early linear filters struggle to balance denoising and image detail preservation. They often destroy lines, blur sharp edges, and reduce fine image details [26]. This drawback is particularly unfavorable in thermal images, where the original physical contours of objects are already fuzzy and difficult to recognize. Secondly, typical linear filters have a fixed frequency response,

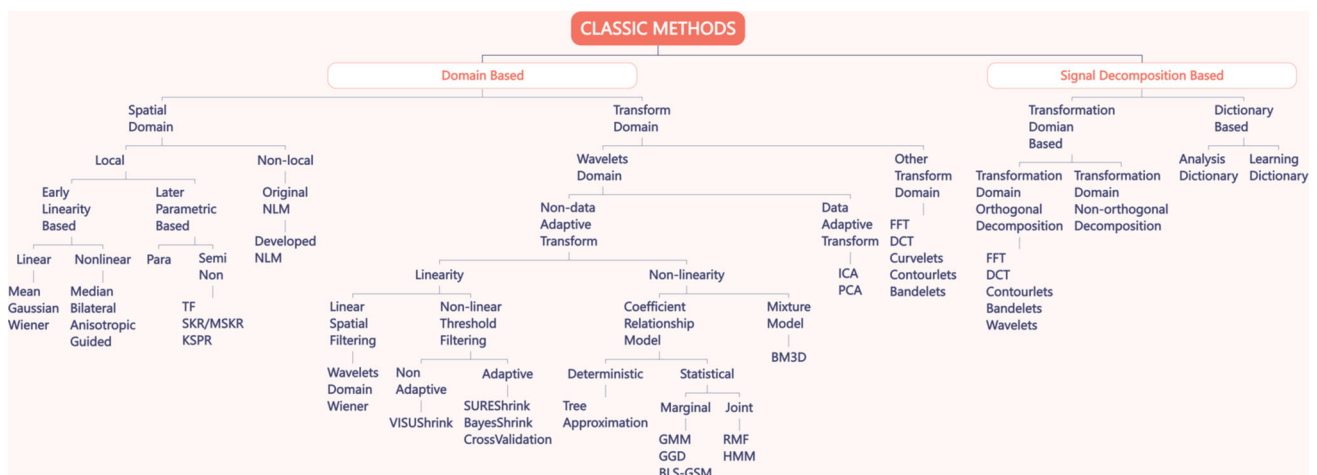


Figure 1 The framework depicts classical methods, covering two perspectives: domain-based approaches and signal sparsity representations.

Table 1 Overview summary of linear filters.

Filter	Mean	Gaussian	Wiener
Principle	Computes the average value of the pixels within a specific window region and assigns this mean value to the pixels located at the center of the window.	Scans each pixel in the image using a user-defined template and replaces the pixel value at the center of the window with a weighted average gray value calculated from the neighboring pixels.	Ensures that the output of the LTI filter closely resembles the original signal.
Key	Average and simple.	The transformation characteristic exhibits a Gaussian decay pattern as the distance from the central pixel increases. The weight assigned to each neighborhood pixel decreases steadily as the distance from the central point increases.	The signal and additive noise are regarded as stationary linear random processes with pre-determined spectral characteristics. The solution sought after should be physically feasible and causal, although the possibility of considering a non-causal solution exists. The primary objective is to minimize the mean squared error.
Merit	Simple and computationally efficient approach. It is particularly effective in handling random interference with approximate positive and negative bias. This method is well-suited for mitigating pulse noise, as the grayscale level of pulse noise typically lacks correlation with the grayscale level of surrounding pixels, and its brightness is significantly higher than that of other pixels.	Achieves outstanding outcomes by considering the distribution of weights based on the distance from the central pixel. Pixels that are farther away from the operator's center are assigned lower weights, which helps preserve the edge characteristics of the image. The Gaussian smoothing parameters employed strike a balance between excessive blurring and insufficient smoothing.	Demonstrates broad applicability, as it can yield favorable outcomes in both continuous and discrete stationary stochastic processes.
Limitation	As the filter radius increases, $A'(i, j)$ progressively becomes more blurred, leading to a reduction in image contrast. The average operation applied during filtering denoises the noise component by spreading it to the surrounding pixels. Although the filtering process reduces the amplitude of noise, it also results in an enlargement of the affected area around the noise pixel.	Increasing the scale factor leads to smoother results, but it can also introduce unexpected distortions to the image. The weight assigned to each pixel is solely determined by the Euclidean distance between the pixels.	Requires generalized and stable input process. It also demands comprehensive knowledge of the statistical characteristics of the input process, which is often unavailable and subject to variation. Moreover, it requires access to all the observed data within a semi-infinite time interval. The complexity of the model is determined by the choice of window size when implementing spatial smoothing.
Improvement	Assigns varying weights to pixels based on the detection of noise pixels through directional differences. It dynamically determines an optimal window size, ensuring a balance between filtering performance and avoiding unintended artifacts. The comparison of noise levels and grayscale values of neighboring pixels is performed using a threshold.		—

lacking the flexibility required to handle signal-dependent noise effectively [27]. Thirdly, the performance of linear filters is highly dependent on the choice of window size, limiting their effectiveness [28].

To overcome these weaknesses, multiple types of modified linear filters were proposed. Combining linear filters with other spatial filters allows further optimization. For example, the mean filter [29] is often combined with non-linear median filters to form hybrid filtering strategies. The application of a parametric extension improves the capabilities of the Gaussian filter [30]. Additionally, the Wiener filter

[31] uses data-driven noise power spectrum estimation for non-parametric denoising. From a theoretical perspective, strategies such as adjustable weights, adaptive window sizes [32], adjustable kernel widths [33], and multi-scale decomposition [34] have significantly alleviated the limitations of early fixed-kernel linear filters, including edge blurring, texture over-smoothing, spatially varying noise suppression, and ringing artifacts in inverse filtering.

In particular, techniques in the transform domain have also been used to achieve better denoising performance [35]. For instance, Mean filters utilize wavelet subbands

Table 2 Overview summary of non-linear filters.

Filter	Median	Bilateral	Guided
Principle	Computes the magnitudes of all vectors within a mask and sorts the pixels based on these magnitudes. It then replaces the value of the target pixel with the value of its neighboring pixel. This process helps eliminate isolated noise points by allowing the surrounding pixel values to approach their true values.	Employs a compromise approach that combines the spatial proximity and similarity of pixel values in the image. Both the spatial and grayscale similarities are taken into account to effectively preserve the image edges.	Incorporates additional information from a supplementary image during filtering process. This supplementary image can be either the original image itself or another input image.
Key	Sorting statistical theory.	Spatially distributed filter based on a Gaussian kernel.	Takes advantage of multi-images.
Merit	Excellent performance in effectively removing typical noise types, such as pulse noise. Images affected by pulse and pretzel noise often exhibit abrupt changes in grayscale values at specific positions. It is crucial to preserve the edges without introducing blurring artifacts. The ideal filter follows the minimum absolute principle, providing a simple and hardware-friendly implementation.	Utilizes a domain filter that calculates filter coefficients based on pixel differences. It applies weighting to average pixels with similar values, and the weight coefficients decrease as the difference in values increases. This approach enables effective filtering of low-frequency information while preserving edges with excellent performance.	Incorporates a booting image to enhance the filtering process by providing additional information. The guided filter effectively preserves edges while denoising, offering a comparable or even superior performance to the bilateral filter. Despite its mixture linearity, the guided filter maintains linear complexity. When calculating a specific output, only the values of the linear function associated with the target point are averaged, even though each pixel is encompassed by multiple windows.
Limitation	The filtering outcome is significantly influenced by the window size, and relying on a manually determined window size can be unreliable. Increasing the window size can lead to image blurring. This approach is not suitable for images containing an abundance of points, lines, or sharp peaks as the filtering effect is compromised when the number of noise pixels exceeds half of the total number of pixels within the window.	Effectively preserves a significant amount of high-frequency information, which cannot be adequately filtered using a bilateral filter in color images. However, caution must be exercised as using the method may lead to excessive smoothing, potentially resulting in the removal of intricate details, leaving only the overall shape intact.	—
Improvement	Excels at preserving image clarity and details. When the center pixel within the window represents a pulse, the filter produces the output of a standard median filter. However, if the center pixel is not a pulse, the filter progressively increases the window size until the noise can be identified as a pulse, and then computes the non-pulse median. This adaptive approach ensures that the filter effectively addresses different types of noise while maintaining the desired level of image quality.	—	—

Table 3 Overview summary of parametric free filters.

Filter	SKR/MSKR	KSPR	TF
Principle	In the context of SKR, emphasis is placed on giving stronger influence to pixels located near an edge, where pixels on the same side of the edge should have a greater impact on the filtering compared to those on different sides. SKR introduces adaption and expansion of the kernel regression concept to address artifact removal, while Maximized SKR aims to enhance SKR by maximizing the metric Q during parameter selection. The metric Q is a non-reference image content metric derived from the SVD of a local image gradient matrix. The process can be summarized in three steps: (1) Detection of the anisotropic patch set in the input noisy image, which serves as the input for calculating the metric Q ; (2) An iterative process to maximize the metric Q using the denoised image as input; (3) Adjustment of parameters in SKR based on the optimization results obtained from the metric Q .	Involves mapping the noisy-clean paired image into a Reproduced Kernel Hilbert Space (RKHS). In this space, the noise-free image is represented as a linear combination of reproduced kernels, following the representer theorem. KSPR employs various kernels to represent different regions within the image. Kernel smoothing is given higher weights in homogeneous regions, while sparse representation is applied to edges in texture regions [70]. This approach ensures that kernel smoothing is effective in preserving image structure, while sparse representation focuses on capturing edge details.	Employs a weighted averaging scheme and incorporates a non-parametric process to calculate weights. These weights are obtained through offline training on a diverse set of images, following a two-step procedure. The first step involves a classification process utilizing adaptive dynamic range coding to assign class indexes to different image patterns. In the second step, the fixed kernel coefficients are replaced with trained coefficients that offer improved adaptation to local image patterns. The filter coefficients are obtained using a Look-Up Table (LUT) generated during the training process. This approach ensures that the filtering process is optimized for specific image patterns, enhancing the overall denoising performance.
Key	Kernel regression.	Kernel regression.	Double processes in denoising, classification and filtering.
Merit	The computational complexity is minimal. It exhibits excellent filtering performance, particularly when dealing with non-Gaussian noise. In comparison to many other features used in denoising algorithms, the Singular Value Decomposition (SVD) is more robust in the presence of noise. This characteristic is crucial for accurately estimating the level of image noise, as the SVD serves as a reliable indicator of texture strength.	Versatile and can be applied to different types of additive noise models, including impulse noise. The formulation of the method, with its semi-parametric nature, enables effective edge preservation capabilities.	The offline training process and the generation of the LUT only need to be performed once, making it efficient and time-saving. This approach can also be applied to address other image enhancement tasks. It takes advantage of sparsity in the dataset, similar to learning-based denoising methods, thereby enhancing the adaptability of local neighborhood filtering.
Limitation	In the presence of numerous homogeneous regions in an image, SKR/MSKR may face challenges in effectively handling the noise due to the high sensitivity of the metric Q to structured regions.	The denoising outcome is typically unsatisfactory when the noise model follows additive Gaussian distribution. The mean gradient-based measurement of smoothness does not yield satisfactory results in the presence of Gaussian noise.	It heavily relies on the accuracy of ADRC. However, it may encounter challenges with severe image degradations, such as high levels of noise.

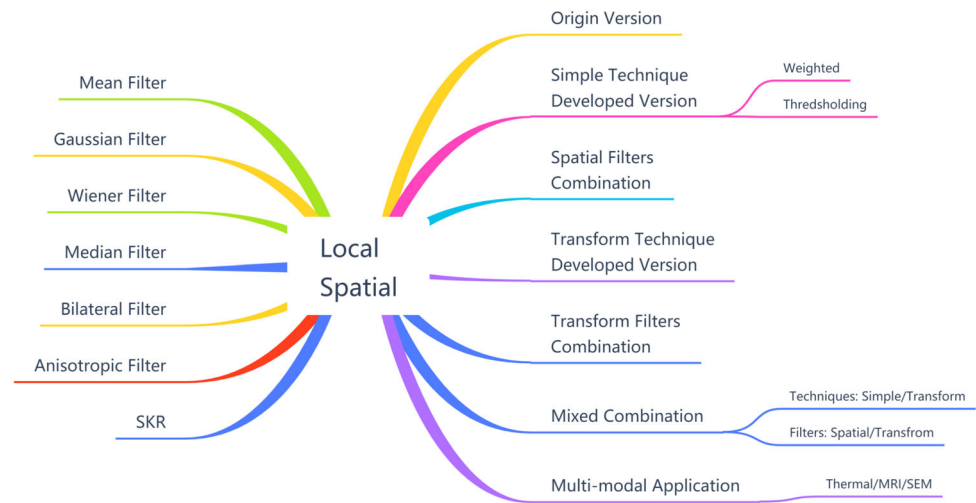
to better preserve edges [36]. Similarly, Gaussian filters have evolved into multi-resolution smoothing via scale-space decompositions [37]. Furthermore, Wiener filters optimize noise suppression by adaptively tuning filter parameters in the transform domain [38]. Moreover, the latest adaptation trends extensively integrate deep learning. Section 5 presents detailed optimization strategies.

In the early stages, to overcome the limitations of linear filtering, researchers investigated a range of non-linear approaches, including median [39], bilateral [40], guided

[41], and anisotropic filters [42]. Nevertheless, non-linear techniques still have shortcomings. The median filter is only effective against high-density salt-and-pepper noise, the bilateral filter often produces “gradient-reversal” artifacts in high-contrast regions, and anisotropic diffusion can introduce a staircase effect on textured areas. Ongoing research therefore focuses on refining non-linear filters.

Through hybrid design, coupling non-linear filters with complementary spatial filters can further enhance their performance. Representative examples include a multi-stage

Figure 2 Framework depicting the longitudinal development process of classical spatial domain-based methods.



hybrid decision-based median-filter framework that integrates additional non-linear operators [43], and an adaptive weighted mean-bilateral composite method [29]. Related work also shows that nonlinear filters can be embedded within parameter-driven models [44] or fused with non-parametric filtering schemes [45], each combination offering distinct gains in robustness and accuracy. Theoretical strategies such as adaptive window sizes, learnable priors, multi-scale decomposition, and recursive temporal filtering have markedly reduced the shortcomings of early fixed-kernel nonlinear filters. For instance, weighted median filters [39] introduce pixel-wise importance factors before applying the median operation, whereas anisotropic diffusion [46] models express the diffusion coefficient as an explicit function of the local gradient magnitude, thus preserving salient edges while suppressing noise.

Meanwhile, transform-domain extensions of non-linear filters exploit frequency representations to enable more adaptive processing. For example, bilateral filtering [47] can apply both spatial and coefficient-value kernels to the transform coefficients. Furthermore, anisotropic diffusion [48] could be reformulated by casting its governing partial-differential equation into the Fourier domain. Similarly, recent developments in non-linear filtering have widely integrated deep learning techniques (see Section 5 for an in-depth discussion). In summary, hybridizing linear filters can greatly enhance the performance of a single linear filter. In addition to hybridization, the integration of parametricity techniques can further amplify performance. Moreover, transform-domain processing and deep learning modules also increase the effectiveness of linear filtering.

Evolution in filtering technology has moved away from linearity towards parametricity [49]. Thus, recent research has focused on parametric, non-parametric, and semi-parametric local filters [50–52]. Early parametric filters share the

philosophy of assuming an explicit noise model and then deriving a locally optimal linear estimator by fitting a small number of parameters to each neighborhood. They exhibit model mismatch, window-size trade-offs, and parameter sensitivity [53]. To improve the performance of parametric filtering, researchers have introduced strategies to collectively mitigate these inherent limitations.

Adjustable weighting [39], adaptive window sizing [54], and multiscale decompositions [55] are popular strategies. From an integration point of view, parametric filters allow further refinement through linearity–parametric composites [56], multiple parametric composites [57], or parametric–non-parametric combinations such as segmentation-guided parametric initialization [58]. In parallel, transform-domain versions of parametric filters use frequency-domain representations to enable more adaptive filtering behavior [59].

Recent research has focused on non-parametric or semi-parametric local filters, which demonstrate impressive denoising capabilities [60]. Unlike parametric techniques examined above, these filters can effectively suppress artifacts during denoising. However, early non-parametric approaches present two inherent drawbacks. First, they generally offer lower statistical power than comparable parametric methods, often requiring larger sample sizes to reach a similar level of significance [61]. Second, because their outputs lack explicit parameter estimates, the resulting findings are less straightforward to interpret and operationalize in practical settings [62].

To overcome these inherent drawbacks, various enhancement strategies have been developed for non-parametric filtering, including adjustable weighting schemes [63] and recursive temporal filtering [64]. Furthermore, hybrid design is a common strategy for improving non-parametric filtering, exemplified by effective linear-non-parametric composites [65]. Specifically, combining mixture linear models

and non-parametric or semi-parametric filters can greatly enhance filter adaptability and flexibility while preserving edges and minimizing denoising artifacts [66]. Moreover, parametric-non-parametric composite filters [67] and non-parametric composites [68] improve filtering performance. These non-parametric filters also have extensions to the transform domain [69], while learning-based methods are also explored.

2.1.2 Non-Local Filter

The previously discussed local filtering methods focused on neighboring pixels within a limited window range, considering only the grayscale value information of the pixels [71]. They did not take into account statistical quantities such as variance or the general distribution properties of pixel values and prior knowledge of noise. To overcome these limitations, the NLM algorithm was then introduced, which uses redundancy in natural images to remove noise. The NLM algorithm was initially proposed as an improvement over Gaussian and median filters, using the observation that using non-local similar patches enhances denoising performance. The original NLM method had a high computational cost as it required searching all pixels in the image to estimate their similarity, as shown in Table 4. Since then, significant improvements have been made to improve the NLM algorithm [72]. These advancements fall into two categories:

- **Computation Acceleration:** Acceleration strategies for the NLM filter aim to reduce its cost without compromising denoising quality. Typical strategies fall into four interconnected categories. The reduction in the search space limits distance evaluations to a subset of

likely similar patches by shrinking the search window [73]. Distance-computation approximations recast the patch-sum-of-squares into constant-time look-ups [74]. Efficient data structures and multiscale schemes replace exhaustive scanning with logarithmic or linear-time high-dimensional filtering [75]. Parallel and learning-based implementations exploit GPUs to compute weights in parallel or predict them directly [76]. In practice, combining two or more of these techniques yields speed-ups from an order of magnitude to real-time performance.

- **Performance Promotion:** Modern efforts to enhance NLM filter denoising quality converge on five complementary lines of research. The refinement of the similarity metric replaces the raw Euclidean patch distance with more robust measures to reduce bias and better preserve fine textures [77]. Weight adaptivity dynamically tunes the bandwidth or exponent of the weighting function according to local noise variance, patch contrast, or Bayesian confidence, often integrating symmetric normalization or probabilistic rescaling to accentuate truly similar neighbors [78]. Multiscale approaches perform coarse-to-fine patch matching in wavelet or pyramid domains [79]. Non-local fusion couples NLM with complementary non-local schemes such as BM3D, kernel regression, and higher-order covariance filters, thus improving both edge retention and texture recovery [80]. Learning-based augmentation focuses on giving NLM with data-driven adaptability while retaining its interpretability. Shallow CNNs or linear-attention blocks can be embedded to predict optimal weights or refine the NLM residual [76].

In general, the NLM algorithm provides a promising approach to noise reduction by effectively utilizing the information present in natural images.

Table 4 Overview of NLM filter.

Metric	Description
Principle	• Assumes similar patches exist in an image. • Identifies regions similar to the target block and then averages these regions. • Patches with similar pixel values to target patches determine the weights.
Key	Execution of the weighted averaging idea in a non-local style.
Merit	• Considers the self-similarity of the entire image. • Uses weighted Euclidean distance to replace Euclidean distance between two patches. • Uses Gaussian distance to substitute Euclidean distance in the weight function.
Limitation	Produces serious artifacts, and performance degrades drastically when the noisy image lacks similar patches.

2.2 Transform Domain

The transform domain method is a well-known and widely used approach for denoising images [81]. It involves converting the image signal into a transform domain and applying the corresponding filters. This method assumes that the noise signal exhibits a distinct regular pattern, allowing easy distinction and separation from the overall transformed signal. The chosen transform largely dictates how the noise and signal components separate. Image-denoising algorithms are commonly classified according to the transform domain in which they operate. In this taxonomy, the algorithms are grouped by the basis used to map the image before attenuation [81]. Various types of transform domain filters have been developed, such as the Discrete Cosine Transform (DCT) [82], wavelets [83], wedgelets [84], curvelets [85], bandlets

[86], and contourlets [87]. In the following, two commonly used transformations are discussed: Wavelet domain and Fast Fourier Transform (FFT) domain.

2.2.1 Wavelet Domain

Wavelet domain-based noise reduction techniques frequently divide image content into multiple sub-bands with different orientations and resolution scales. This transformation allows for the separation of signal from noise. The significant signal, characterized by large coefficients in the low-frequency range of the wavelet domain, is isolated, while noise and finer details reside in the high-frequency sub-bands. Wavelet-based methods fall into non-adaptive and adaptive categories, depending on the basis function's characteristics. Non-adaptive wavelet pipelines have been revitalized with learnable coefficient-modulation blocks. For instance, Li et al. [88] presents a method that embeds a discrete wavelet analysis-synthesis pair inside a lightweight Vision Transformer and trains small MLP gates that rescale each sub-band. Adaptive wavelet approaches learn either the basis itself or data-dependent thresholds. For instance, Tian et al. [89] proposes a multi-stage scheme that first trains a sparsifying, orientation-adaptive lifting wavelet.

Wavelet paradigm has evolved from handcrafted coefficient thresholding to adaptive, learnable, and context-aware multi-resolution filtering while still preserving the interpretability and efficiency that make wavelets attractive for practical denoising. Recent approaches for optimizing wavelet-domain denoising fall into five logically organized stages, from “still purely analytic” up to “fully learnable”. Enhanced fixed-basis shrinkage designs smarter, analytic threshold or shrinkage rules [90]. Multi-scale hybrid priors couple wavelet shrinkage with other sparsity mechanisms. Data-adaptive wavelet bases learn lifting-scheme filters or oriented wavelet packets from data [91]. Sub-band aware gating inserts lightweight MLP gates or attention blocks between analysis and synthesis to modulate each sub-band [92]. Wavelet front-ends for deep priors treat the wavelet transform as a fixed or learnable encoder [93].

2.2.2 FFT Domain

Numerous investigations explore denoising techniques in the FFT domain. For example, Zhang and Jiang [94] presents a basic denoising concept utilizing the Fourier transform, where the noisy signal undergoes an initial FFT, followed by applying a Butterworth low-pass filter to eliminate noise frequencies. As an attractive denoising method, it converts convolution into simple, element-wise multiplication, turning costly spatial filtering into simple operations. FFT also provides global context while preserving phase relationships,

avoiding the edge blurring and ringing often introduced by strictly local filters, thus making FFT a core component of both classical and modern pipelines.

FFT domain-based denoising has evolved from static low-pass rejection to adaptive, statistically informed filtering and finally to learnable spectral-spatial networks. Early Fourier domain methods [95] treated noise as purely high-frequency energy and suppressed it with ideal low-pass filters after a forward FFT. To overcome this rigidity, recent approaches estimate the spectral characteristics adaptively. Typical schemes include: (1) measuring the local power spectral density to choose a data-dependent cutoff [96], (2) performing spectral subtraction by modeling noise as an additive spectrum and subtracting it bin-wise [97], or (3) applying Wiener-like gains that weight each frequency according to an estimated signal-to-noise ratio [98]. The latest trend embeds FFT [99] layers within neural architectures, allowing end-to-end filtering optimization. These models can learn task-specific spectral masks that suppress noise while selectively increasing informative high-frequency components, achieving state-of-the-art results on natural images.

3 Signal Decomposition-Based Methods

Transform domain denoising methods also allow for sparse signal representation [100]. When considering sparse decomposition and representation, a noisy image allows division into two components: the actual signal and the noise. The real signal being structured, is considered the sparse component, while noise is considered random, uncorrelated, and lacking structure. Setting a sparse representation threshold allows the real signal to be preserved and a clean structured signal from the original image is extracted, while eliminating random noise. Sparse signal representation aims to represent signals with minimal elements in an ultra-complete dictionary, achieving a more concise representation [101]. The key objectives of sparse signal representation include decomposing signals sparsely and generating dictionaries. According to sparse decomposition theory, the sparser the signal representation, the higher the accuracy of the reconstructed signal.

The transform domain-based methods discussed above, in terms of signal sparsity, fall into a type of early signal decomposition known as orthogonality. Orthogonal and double-orthogonal dictionaries [102] saw wide use in the transform domain prior to the introduction of sparse representation theory. Examples include the wavelet domain and the Block Matching and 3D Filtering (BM3D) model [103]. However, these methods are limited by poor adaptability, as they cannot flexibly and comprehensively represent signals. To address this drawback and enhance the adaptiveness and sparsity of the decomposed signal, wavelet-domain non-

orthogonal decomposition was developed. Non-orthogonal multiscale representations alleviate the “basis mismatch” problem by introducing redundancy and shift invariance, yet the underlying atoms remain hand-designed. Consequently, their sparsifying power remains limited when faced with highly unstructured or complex textures.

To achieve greater sparsity, researchers turned to dictionary-based representations. The earliest analysis dictionaries extend redundant bases into fully overcomplete dictionaries that forgo orthogonality and require only that the signal admit a sparse set of coefficients within the dictionary. Subsequently, dictionary learning-based sparse representation methods have gained popularity due to their improved effectiveness and adaptability, further advancing wavelet analysis-based sparse representation. The use of a learned dictionary simplifies the extraction of signal information and makes signal processing more convenient. Figure 3 illustrates the evolution of classical signal decomposition-based classification methods.

3.1 Transformation Domain-Based

3.2 Orthogonal Decomposition

Sparse-signal representation provides a lens through which to understand denoising techniques applied in the aforementioned transform domains. Transforming signals into transform domains can be seen as signal decomposition. Transform-based signal decomposition methods fall into two types based on the orthogonality of their decomposition basis functions: transform domain orthogonal decomposition and transform domain non-orthogonal decomposition. Well-known algorithms such as the DCT, wavelets, wedgelets, curvelets, bandlets, and contourlets fall primarily under the category of transform domain orthogonal decomposition.

Recently, orthogonal-decomposition denoisers have evolved from pure coefficient thresholding to adaptive, physics-

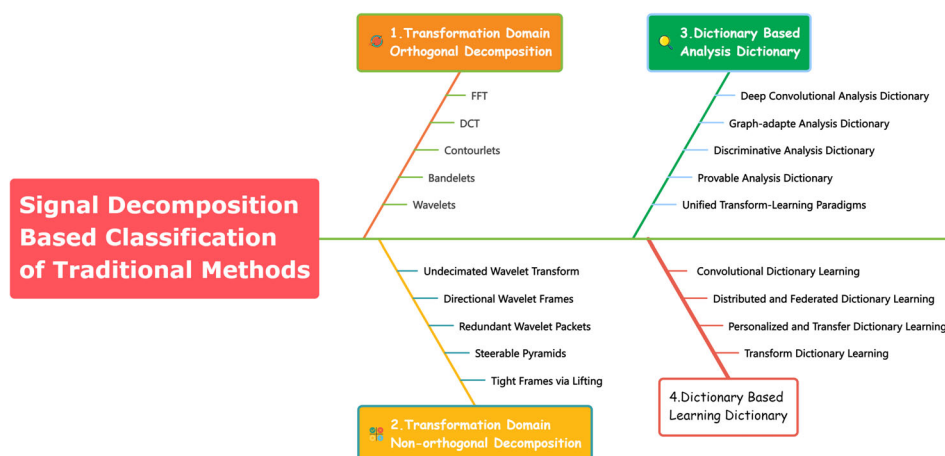
aware, and deep-integrated frameworks that retain the theoretical elegance of orthogonality while matching the performance of more flexible non-orthogonal methods. Advances in orthogonal-decomposition denoising converge on four interrelated directions. First, the field has shifted from fixed to adaptive atoms: orientation-adaptive bandlets and graph wedgelets dynamically rotate or reshape canonical basis functions so that they align with local anisotropic structures, capturing details that conventional Daubechies or contourlet atoms often miss [84].

Second, a “deep-hybrid” renaissance has emerged: fixed FFT [104], DCT [105], wavelet [83], or contourlet [106] encoders are coupled to the lightweight CNN or Transformer back-ends, combining the interpretability and energy fraction of the orthogonal transforms with data-driven adaptability. Third, orthogonality sees increasing domain-adapting: Each basis is paired with bespoke thresholding or fusion strategies that match the noise statistics of the target data. For example, curvelets are preferred in biomedical microscopy [85], and block-DCT remains indispensable for compression-oriented pipelines [107]. Finally, real-time deployment has become feasible: the separable structure of FFT or DCT kernels and lifting-scheme wavelets maps efficiently to SIMD engines and FPGA fabrics [108], making orthogonal decompositions a practical choice for on-device denoising under strict latency and power constraints.

3.3 Non-Orthogonal Decomposition

Non-orthogonal decomposition, characterized by redundancy, denotes any linear transform that maps an N -sample signal to a coefficient vector of length $M > N$ using non-mutually orthogonal basis atoms. Because the atoms overlap in both space and frequency, the transform does not form an orthonormal basis but rather an overcomplete frame in the sense of frame theory. The accompanying redundancy confers two principal advantages. First, this redundancy achieves

Figure 3 Illustrating the progression of classical signal decomposition-based denoising techniques, emphasizing the representation of signal sparsity. The framework includes three distinct stages: an initial phase based on transforms, followed by an analysis dictionary-based phase, and finally, a phase using learning dictionaries.



shift and rotation invariance because it represents each spatial location at every scale and orientation, eliminating pseudo-Gibbs artifacts. Second, the additional degrees of freedom allow the atoms to better align with highly textured structures. Energy that would otherwise scatter across many orthogonal coefficients is concentrated into a few large frame coefficients, yielding a sparser representation. Non-orthogonal, multi-scale decompositions can be grouped into several well-defined families, each characterized by a particular way of injecting redundancy:

- **Shift-invariant or undecimated wavelet transforms (UDWT/SWT)** eliminate the down-sampling step of the DWT. Every sub-band therefore retains full spatial resolution, endowing the representation with translation invariance. Research on UDWT/SWT has advanced in several complementary directions. 1) Redundancy management saw refinement through tight-frame constructions such as the undecimated dual-tree complex wavelet transform and non-subsampled contourlet or shearlet frame. For example, Liu et al. [109] implements an undecimated dual-tree CWT with limited redundancy while retaining directional selectivity for speckle removal. 2) Adaptive coefficient processing exploits the full-resolution sub-bands. Recent work derives edge-aware or cone-of-influence masks and employs data-driven shrinkage such as trimmed-mean, Teager-energy or Bayesian thresholds. Veerakumar et al. [110] demonstrated a novel denoising technique based on UDWT, combining the Cone of Influence (COI) analysis technique with UDWT to effectively remove impulse noise while preserving edges. 3) Hybrid and learning-augmented architectures focus on coupling this analytic front-end with lightweight learning blocks [111]. 4) Hardware-oriented implementations [112] map the spatially invariant à-trous convolutions onto GPUs and FPGAs.
- **Directional wavelet frames**, including contourlets, curvelets, shearlets, and other directional filter-bank schemes, augment the radial scaling of classical wavelets with angular partitioning. The resulting coefficients are highly sparse for images dominated by curvilinear structure. Since their introduction in their early stages, they have evolved along four complementary research lines. 1) Shift-invariant, tight-frame refinements suppress down-sampling and enforce tight-frame conditions, producing non-subsampled contourlet, undecimated curvelet, and shift-invariant shearlet transforms [113] whose coefficients remain stable under pixel shifts. 2) Higher angular resolution and adaptive orientation replaces fixed dyadic angular partitions [114]. Flexible fan filter banks, adaptive shearlet pyramids, and orientation-steered bandlet or wedgelet frames allocate more directions where edge curvature is high. Energy is concentrated into fewer, larger coefficients, increasing sparsity on curvilinear features. 3) Hybrid and learning-augmented architectures routinely outperform complex U-Nets while using a fraction of the parameters and retaining interpretability. Directional encoders now see embedding in lightweight CNN or Transformer back-ends. Frozen contourlet [115] or shearlet analysis layers feed sub-band magnitudes to MLP gates or self-attention blocks that learn content-adaptive shrinkage. 4) Domain-specific extensions are developed and paired with bespoke statistical priors. For example, curvelets are increasingly applied to biomedical microscopy [116] and seismic volumes, bandlets to remote-sensing pansharpening, and shearlets to low-dose CT. Meanwhile, hardware deployment prompts the separable filtering structure of directional frames mapping efficiently to GPUs and FPGA fabrics.
- **Redundant wavelet packets and framelets** generalize the analysis tree by iteratively splitting not only approximation but also detail bands, producing a rich library of time-frequency tilings. In the context of wavelet families, a basis set dictionary consisting of wavelet packets associated with the successive decomposition of approximation and detail coefficients was developed. This approach yields a highly redundant representation of the signal. Over the past decade redundant wavelet packet and framelet schemes have evolved from static “best-basis” libraries into highly adaptive, task-specific tilings. 1) Adaptive packet trees: Instead of the classical dyadic binary tree, modern algorithms grow anisotropic or fully balanced trees and prune them by Bayesian risk [117] or minimum-description-length criteria. 2) Dual-tree and undecimated variants: To add shift invariance and orientation selectivity, researchers now combine packet splitting with the dual-tree complex wavelet transform [118]. 3) Learned framelets and packet dictionaries: Convolutional-framelet networks treat the analysis [119] filters as learnable parameters constrained by tight-frame conditions, allowing end-to-end training while preserving the analytic inverse. 4) Task-oriented anisotropic packets: For data with strong directional features, packet trees are built on curvelet [120] or shearlet atoms. 5) Real-time implementations [121]: GPUs and FPGA fabrics consider adaptivity.
- **Scale-space or steerable pyramids** implement a cascaded sequence of low-pass and band-pass filters whose impulse responses are designed to be steerable. They achieve both orientation flexibility and a tight frame, making coefficient manipulation and reconstruction straightforward. Since the original steerable pyramid, research on scale-space or steerable pyramids has advanced along intertwined tracks. 1) Broader steerability and tighter frames. Modern designs build analytic impulse responses

with shift-invariant phase, permitting arbitrary rotation in polar decomposition [122]. 2) Data-adaptive and learned steering. Recent “dynamic” pyramids estimate a local orientation map and rotate the basis functions dynamically, concentrating curvilinear energy into fewer coefficients. At the same time, the analysis filters begin to be treated as learnable parameters [123] constrained by steerability polynomials. 3) Domain-specific deployments. Application-oriented variants pair the tight-frame property with priors matched to imaging noise statistics. For example, complex steerable pyramids for retinal vessel enhancement [124], and hyperspherical steerable frames for spherical panoramas. 4) Real-time exploration. The convolutions of steerable filters [125] map efficiently to GPUs and FPGAs due to their separable radius and angle structure.

- **Tight frames derived via lifting schemes** construct pairs of biorthogonal analysis and synthesis filters that may overlap spatially while still guaranteeing perfect reconstruction. These structures have undergone several key developments: 1) Generalized lifting for arbitrary domains. Early work extended the classical lifting steps into a generalized lifting scheme that factors any biorthogonal or tight frame into elementary filters [126]. 2) Directional lifting transforms. To inject orientation selectivity into tight frames, non-subsampled lifting structures were devised that maintain perfect reconstruction while including directional prediction and update steps [127]. 3) Smooth high-vanishing-moment framelets. Lifting-based designs construct smooth framelets [128] with arbitrarily many vanishing moments by iterating oversampled filter banks. 4) Learnable lifting and convolutional framelets. More recently, the lifting filters themselves have become learnable under tight-frame constraints [129]. Convolutional-framelet networks train both prediction and update filters end-to-end. 5) Hardware acceleration. Since lifting steps are simple local operations, they map efficiently onto SIMD architectures, GPUs, and FPGA fabrics [130].

3.4 Dictionary-Based

3.4.1 Analysis Dictionary

Analysis-dictionary methods have evolved from analytic, hand-created transforms to fully data-driven, end-to-end learned operators. The seminal K-Means Singular Value Decomposition (K-SVD) algorithm first demonstrated how to learn an overcomplete analysis dictionary by alternating sparse coding with a joint update of the analysis atoms, establishing dictionary learning in the context of the “analytic sparse model”.

K-SVD employs multiple similar dictionary atoms to reconstruct individual pixels. K-SVD updates the dictionary during the approximation process based on information from the denoised image. This approach successfully improves image representation by using a more adaptable model and introducing the concept of learning through examples. However, the lack of structure in the dictionary leads to increased computational complexity. Although a redundant dictionary is used to represent each patch, finding a suitable match in the dictionary remains challenging because dictionary atoms can be relatively diverse, even with slight changes in the input patch. The iterative optimization process of K-SVD involves two key steps:

- **Sparse Coding:** The initial dictionary is used to approximate all patches sparsely.
- **Dictionary Update:** The dictionary is updated to enhance the quality of the sparse approximation.

Recent advances in analysis-dictionary learning have centered on extending the classical K-SVD framework into highly adaptive, domain-aware, and deep-integrated forms: 1) *Deep Convolutional Analysis Dictionaries (DeepCAM)*: Huang and Dragotti [131] introduced the DeepCAM, which parameterizes the analysis filters as convolutional kernels learned end-to-end under tight-frame constraints. 2) *Graph-adapted Analysis Dictionaries*: Li et al. [132] proposed signal-adapted tight frames on graphs by generalizing the lifting scheme to arbitrary graph domains. Analysis atoms are trained to match the graph Laplacian spectrum, producing over-complete dictionaries. 3) *Discriminative Analysis Dictionary Learning (SDADL)*: Du et al. [133] developed a SDADL method, which alternates analysis-sparse coding with a joint update of atoms and a discriminative projection. 4) *Provable Analysis Dictionary*: Li et al. [134] offers theoretical guarantees by alternating convex sparse-coding steps with closed-form analysis-atom updates, extending K-SVD’s alternating minimization into globally convergent regimes. 5) *Unified Transform-Learning Paradigms*: Peng [135] shows how alternating sparse coding and transform updates can be viewed as a unified framework including analytic bases, learned transforms, and unrolled neural networks.

3.4.2 Learning Dictionary

Dictionary learning refers to the data-driven construction of overcomplete bases in which signals admit sparse representations. Early methods alternated between sparse coding and atom updates, learning a single, global dictionary suitable for a broad class of signals. To handle large datasets, Stochastic Gradient Descent and Online Dictionary Learning (ODL) approaches update the dictionary incrementally as new samples arrive, enabling real-time adaptation and reducing

memory footprint [136]. Dictionary learning has transitioned from offline, monolithic bases to real-time, structured, and learnable transforms:

- **Convolutional Dictionary Learning (CDL)** extends patch-based DL by learning small filters applied across the entire image, yielding translation-invariant sparse codes. Yin et al. [137] proposes a deep CDL framework, where image patches drive a two-stage pipeline—non-local sparse coding followed by a small CNN for global similarity matching—delivering high-fidelity denoising via end-to-end learned convolutional filters.
- **Distributed and Federated Dictionary Learning** parallelizes local sparse coding across multiple nodes while synchronously updating a shared dictionary. Lillelund et al. [138] introduces Cloud K-SVD, which trains a mutual dictionary across Kubernetes-deployed nodes, demonstrating that overcomplete dictionaries can be learned collaboratively in the cloud.
- **Personalized and Transfer Dictionary Learning** adapt sparse-representation models to heterogeneous data by jointly optimizing shared and private atoms for each user or domain. Liang et al. [139] defines Personalized Dictionary Learning (PerDL), splitting global and local atoms to adapt to heterogeneous imaging sites, and jointly optimizes shared and private dictionaries for robust denoising.
- **Transform Dictionary Learning**, instead of synthesizing signals from sparse atoms, seeks operators under which signals produce sparse analysis coefficients. Zhang et al. [140] presents a comprehensive transform-learning paradigm that alternates sparse-coding in an analysis model with closed-form transform updates, unifying analytic bases, learned transforms, and unrolled neural architectures.

Recent advances include group structures, discriminative priors, and domain adaptation in dictionary learning. Janjušević et al. [141] unrolls a group-sparse convolutional dictionary learning algorithm into a non-local neural network, where sliding-window sparse attention arises naturally from a weighted group. Graph and manifold extensions also contribute to sparse representations of signals on social networks, sensor meshes, and 3D surfaces. Shi et al. [142] generalizes lifting-scheme dictionaries to arbitrary graph domains, constructing signal-adapted tight frames on graphs.

4 Denoising of Thermal Images

Denoising methods were investigated on thermal image datasets for comparison. Evaluating the denoising performance

of classical methods on thermal images using simulation and real tests.

4.1 Simulation Test

The simulation test evaluated the denoising performance. Transforming RGB images generated simulated thermal images. Recreating the Near Infrared (NIR) channel from RGB images via image-to-image translation used a supervised approach. Image-to-Image Translation (I2I) entails transferring images from a source domain (A) to a target domain (B) while preserving essential content representations. Several models exist for supervised I2I translation [143–146]. Its compact network design led to the use of the Conditional Generative Adversarial Network (cGAN) architecture [143]. The generator component used the UNet architecture, a well-established framework for image segmentation [147], while PatchGAN served as a discriminator [143, 148]. The original cGAN framework [143] employed a combination of the L1 distance metric and the GAN objective function to enhance realistic image generation. Figure 5 shows the simulated thermal image and the corresponding RGB pairs.

The addition of Gaussian noise created a set of noisy thermal images for testing. The integration of Gaussian noise involved directly overlaying a noise matrix of the same size as the original image. A dataset specifically constructed for this evaluation used several classical denoising methods, including Mean, bilateral, NLM, guided, BM3D, and median filters. The window size for all filters was configured to 4×4 pixels. For the NLM and BM3D algorithms, the template size was 8×8 pixels, and the search window size was 16×16 pixels. In addition, the guided and bilateral filters used a radius of $r = 2$ and $\eta = 0.01$.

Quantitative evaluation used the objective indicators PSNR and SSIM, commonly used in image denoising. Table 5 presents the average PSNR and SSIM values from the simulation results in the constructed infrared image test set. Figure 4 shows the denoising contrast results, presenting the denoised thermal images which comprised grayscale images with dimensions of 256×256 .

Generally, increasing noise level adversely affects image quality, accompanied by decreasing PSNR and SSIM values. Noise introduces additional random interference, damaging image details and structural information, consequently impacting denoising outcomes. Specifically, as the noise level increases, the mean squared error of the degraded image increases, resulting in decreasing PSNR values. In addition, the SSIM evaluation metric accounts for brightness, contrast, and structural differences in images. Therefore, as noise levels increase, they compromise structural information, leading

Table 5 Average PSNR and SSIM values on the constructed image set at various noise levels.

Type	Method	$\sigma = 15$	$\sigma = 25$	$\sigma = 50$
PSNR	Mean[149]	34.25	33.48	31.26
	Medium [149]	34.91	33.92	32.75
	Bilateral [150]	35.72	34.87	34.16
	BM3D [103]	35.48	36.71	36.42
	Guided [151]	35.63	34.59	33.94
	NLM [18]	35.56	36.68	36.27
SSIM	Mean	0.727	0.614	0.582
	Medium	0.763	0.645	0.617
	Bilateral	0.864	0.793	0.695
	BM3D	0.778	0.883	0.826
	Guided	0.851	0.764	0.673
	NLM	0.784	0.849	0.812

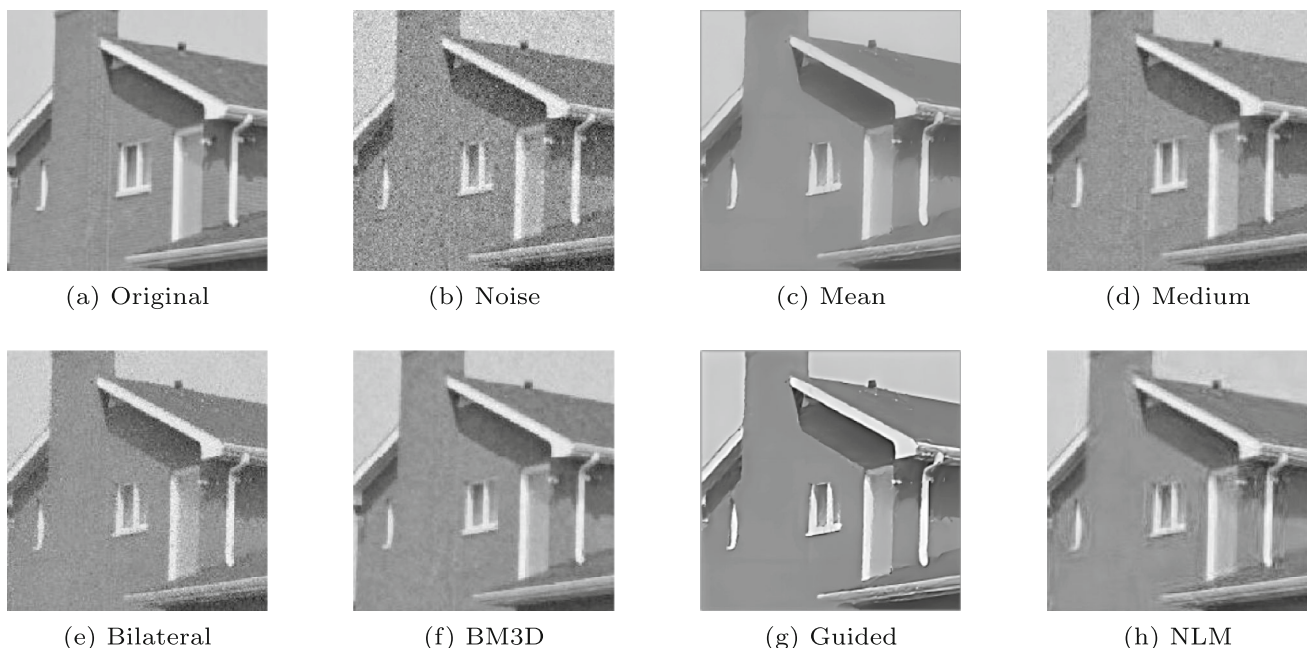
to decreasing SSIM values. However, BM3D and NLM show increases in PSNR and SSIM at $\sigma = 25$ and $\sigma = 50$ compared to $\sigma = 15$. In conditions of very low noise, the performance of BM3D decreases. Its collaborative-thresholding and block-matching algorithms are optimized for moderate to high noise levels. However, with very low noise, these methods can misinterpret subtle genuine textures as noise, resulting in loss of detail and blocky artifacts (Fig. 5).

NLM also shows mild over-smoothing in such low-noise regimes due to its strict patch-similarity thresholds, although the effect is less pronounced than with BM3D. In contrast, adjustable parameters and strong edge-preservation

characteristics of bilateral and guided filters can better retain fine details at low noise levels. Specifically, when noise increases to $\sigma = 25$, PSNR and SSIM values for both BM3D and NLM decrease again as noise increases. The experimental data reveal that each algorithm effectively improved the PSNR and SSIM of the image at different noise levels. In particular, the mean method exhibited the lowest PSNR and SSIM values compared to the other methods at the same noise level. However, the NLM and BM3D methods achieved relatively high PSNR and SSIM values. Furthermore, BM3D and NLM methods showed minimal differences in PSNR and SSIM, indicating their high effectiveness in denoising infrared images.

4.2 Real Test

The real test evaluated denoising performance. The real test data originate from real thermal image datasets, including the FLIR [152] and the OSU thermal pedestrian database [153], which are benchmark thermal datasets. Video frames from the FLIR dataset recorded at 30 fps, whereas image sequences sampled at 1 or 2 fps; the dataset includes over 14,000 images and numerous classes (e.g., person, car, bicycle, dog). The OSU thermal pedestrian database captured data using the Raytheon 300D thermal sensor core at a sampling rate of less than 30 Hz; it contains 284 frames. The datasets comprise two categories: gray-noisy and color-noisy images. Gray-noisy images refer to noisy thermal images in grayscale; color-noisy images refer to thermal images with color channels.

**Figure 4** Simulation results on the house image under different classical methods.

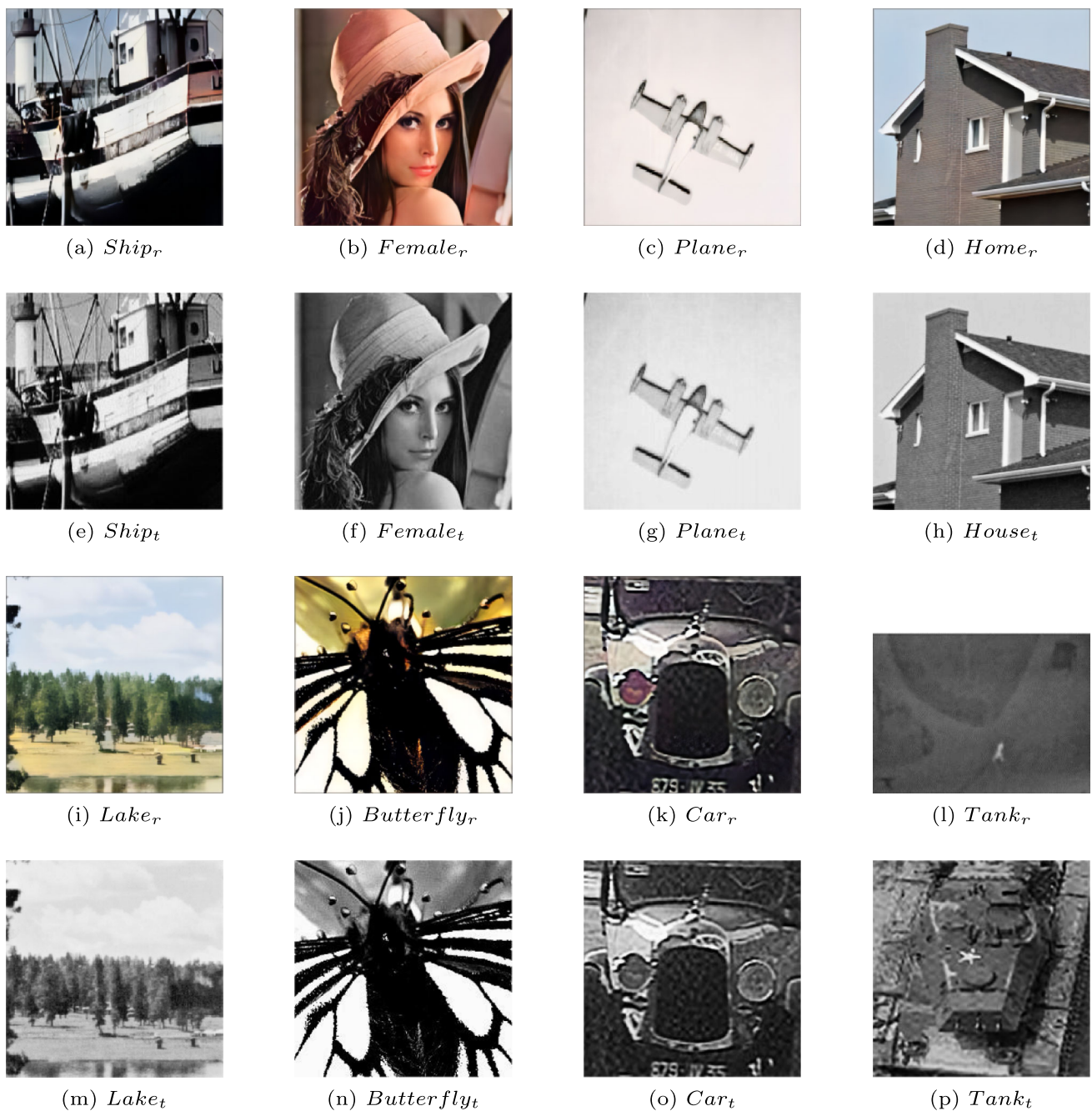


Figure 5 Simulated thermal image and corresponding RGB pairs for simulation test. Labels r denote original RGB images selected from open RGB benchmarks, and labels t denote transferred thermal images. The cGAN [143] performs this image translation.

Adding additive Gaussian noise to the original image created noisy images. The evaluation used three noise levels: $\sigma = 15, 25, 50$. The real test applied the same classical denoising methods as the simulation test, including Mean, bilateral, NLM, guided, BM3D, and median filters, using the same parameter configurations (e.g., window size 4×4 , NLM/BM3D template 8×8 and search 16×16 , guided/bilateral radius $r = 2$ and $\eta = 0.01$).

Table 6 shows the PSNR and SSIM values of the real test under additive Gaussian noise conditions at $\sigma = 25$. All filters reduce noise in thermal images, improving visual quality. However, the denoising performance on real thermal images was worse than on simulated thermal images at the same noise level. Real thermal images showed relatively low average PSNR and SSIM values.

Table 6 Average PSNR and SSIM values on both color-noisy and gray-noisy images for additive white noise at $\sigma = 25$.

Dataset	Methods	Channel	PSNR	SSIM
OSU	Mean	color	23.47	0.529
		gray	23.31	0.517
	Medium	color	23.93	0.548
		gray	23.84	0.533
	Bilateral	color	24.86	0.695
		gray	24.73	0.676
	BM3D	color	26.66	0.791
		gray	26.59	0.762
	Guided	color	24.62	0.653
		gray	24.58	0.629
	NLM	color	26.71	0.743
		gray	26.69	0.712
FLIR	Mean	color	23.42	0.521
		gray	23.36	0.509
	Medium	color	23.78	0.538
		gray	23.65	0.527
	Bilateral	color	24.77	0.682
		gray	24.62	0.663
	BM3D	color	26.58	0.775
		gray	26.47	0.764
	Guided	color	24.59	0.649
		gray	24.54	0.618
	NLM	color	26.61	0.736
		gray	26.53	0.705

Results show higher average PSNR and SSIM on color-noisy images than on corresponding gray-noisy counterparts. Adding the same noise variance independently to R, G, and B results in the same per-channel MSE as in the grayscale case. Averaging the overall MSE over three channels lowers the effective distortion per channel in color-noisy images; a smaller MSE yields a higher PSNR. SSIM shows the highest sensitivity to changes in luminance and structure, while showing little sensitivity to uniform biases in hue and saturation. In color-noisy images, a substantial fraction of the noise energy falls into the chrominance (Cb or Cr) channels. In contrast, grayscale noise perturbs the only available channel, the luminance. Therefore, SSIM falls more sharply in gray-noisy images since chroma noise is less “visible” to the structural similarity metric.

Figures 6 and 7 show qualitative evaluations of real test recovery results. The figures also note PSNR values for different methods. Most filters do not perform efficiently in real thermal image denoising because they cannot reconstruct fine texture details or maintain the high-frequency component of frames. However, frames show clearer results compared to the original images.

5 Modern Denoising

The preceding sections surveyed classical denoising methods, organizing them by domain-based and signal-decomposition-based taxonomies, presenting their theoretical foundations and representative algorithms, and introducing the most recent refinements within each category. This section focuses on integrating these classical techniques with modern deep-learning frameworks. This section first examines how spatial, transform domain-based and dictionary-based denoisers are embedded as explicit modules within convolutional and attention-based architectures. The section then analyzes, as a case study in multimodal denoising development, the performance of such hybrid approaches on thermal imaging, highlighting their ability to preserve fine texture details and structural fidelity. Finally, the section identifies several promising avenues for future research, ranging from learnable tight-frame transforms and cross-domain adaptation to hardware-aware network designs, that promise to reconcile interpretability, efficiency, and robustness in next-generation image restoration.

5.1 Spatial Domain

Linearity-based filters are encapsulated in differentiable modules, allowing direct integration into deep neural networks. Adaptive mean filtering utilizes DFN [154] to generate local convolution kernels dynamically according to input features, enabling mean or box filters with variable sizes and weights. Lightweight subnetworks predict the standard deviation or kernel size of Gaussian filters [155], adaptively adjusting the blurring level based on local noise and texture complexity. Traditional Wiener filtering embeds into an FFT layer [156], where the network estimates the signal-to-noise ratio for each frequency component via regression or classification branches, subsequently weighting the spectrum according to the Wiener gain formula. Introducing a Soft-Median function [157] overcomes the non-differentiability of traditional median operations, replacing the selection of the window’s median with a smooth, weighted averaging operation, thereby facilitating gradient propagation and end-to-end training. Deep Bilateral Learning maps bilateral filtering to a low-resolution bilateral grid [158], predicting local affine transformation coefficients through CNN branches, and reconstructing the full-resolution result using differentiable slicing operations. Tensor-guided anisotropic PDE methods [159] estimate local structure tensors and automatically adjust diffusion intensities, allowing spatially varying anisotropic diffusion.

Parametric-based filters use differentiable kernel spline layers to achieve kernel spline regression [160] through backpropagation-compatible convolution operations. These layers parameterize control point positions and polynomial

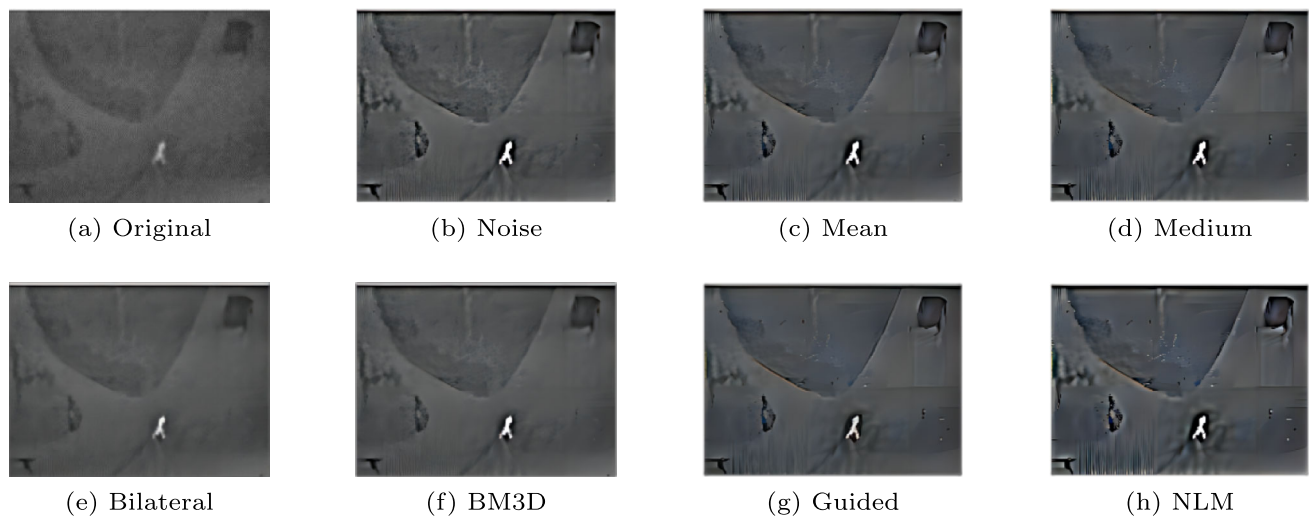


Figure 6 Real test results on a gray-noisy image under different classical methods.

orders, facilitating the simultaneous learning of filter locations and kernel shapes during end-to-end training. Dynamic kernel prediction [161] formulates the Nadaraya-Watson kernel regression as differentiable convolutional layers; lightweight CNN branches adaptively predict kernel bandwidths and weights in real time. Algorithm unrolling maps iterative solutions of total variation [162] or graph trend filtering into neural network layers, with each filtering iteration corresponding to a learnable linear transformation and non-linear gating, thus automatically optimizing step size and threshold parameters during training. Multi-scale fusion aggregates kernel regression outputs across different scales via pyramid structures or parallel branches [163], subsequently using attention mechanisms or gating networks.

In recent years, NLM has transitioned from an isolated post-processing step to being natively integrated within deep neural networks as differentiable “non-local” modules. Some approaches unfold the iterative weight-updating procedure of NLM into a sequence of neural layers [164], embedding learnable parameters in the weighting functions for end-to-end optimization. To adapt to varying noise levels and resolutions, recent research [165] often inserts multiscale non-local

modules into the CNN backbone, running parallel with local convolutional or deformable convolutional layers, thus producing local-non-local hybrid features that simultaneously preserve local details and enforce global consistency. Other strategies include: 1) replacing classical similarity computations with linear attention mechanisms; 2) integrating non-local blocks with self-attention modules; and 3) employing learnable self-similarity priors in the feature domain.

Deep learning-based classical spatial-domain filters show particular effectiveness for infrared image denoising. Infrared image noise often manifests as impulsive or speckle patterns, making it difficult for standard convolutions to remove noise without blurring thermal texture boundaries. Embedding median and bilateral filters as front-end layers enables precise suppression of isolated noise spikes and intensity fluctuations while maintaining the clarity of interfaces between different temperature regions, thus providing cleaner and more structured feature maps for subsequent CNN processing. For example, Liu et al. [166] integrated an initial trainable bilateral filtering layer whose parameters undergo adversarial fine-tuning, thus preserving edges before deeper restoration stages. Hoorfar et al. [167] proposed a method that inserts

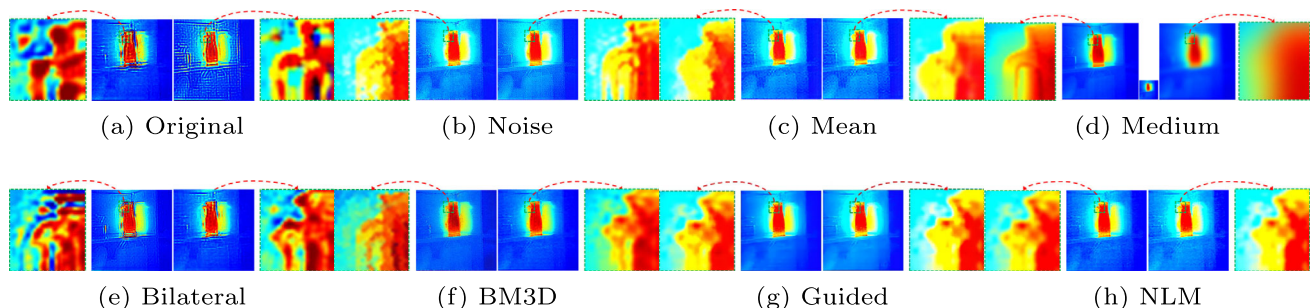


Figure 7 Real test results on a color-noisy image under different classical methods.

mean, median, Gaussian and bilateral filters as front-end layers before a U-Net; median filtering yielded the largest IoU improvement in hot flush detection.

Utilizing parallel filters at the same feature-map level enables the fusion of their outputs through adaptable convolutional or attention-based weighting methods. This approach accommodates diverse noise types and adaptively selects the optimal filter combination for each input, improving generalization. Furthermore, Li et al. [168] introduces a dynamic channel attention block that adaptively modulates Gaussian-like convolution kernels based on local noise statistics, significantly increasing denoising performance in low-contrast thermal scenes. After front-end spatial filtering removes the bulk of local noise, integrating subsequent channel or spatial attention—and even non-local—modules captures long-range dependencies and thermal patterns, further suppressing residual noise and reinforcing texture consistency.

In summary, classical spatial domain filters integrated into deep learning frameworks retain interpretability and computational efficiency while gaining data-driven flexibility, significantly improving the performance of denoising and image enhancement tasks. The main applied techniques include:

- **Differentiable Approximation:** Methods like Soft-Median or piecewise-linear derivatives smoothly approximate “hard” operations such as sorting or partial derivative steps of PDE, ensuring end-to-end trainability.
- **Dynamic Parameter Prediction:** Subnetwork branches dynamically generate filtering kernel parameters based on local features in real-time, enhancing adaptivity.
- **Modular Design:** Decomposing traditional filtering formulas into sequences of convolution, multiplication-addition, and mapping operations encapsulates them as standard neural network layers.
- **Network Unrolling:** Unfolding iterative weight-updating procedures into a sequence of neural network layers makes each layer correspond to one step of patch matching and reweighting.
- **Multi-scale Fusion:** Inserting multiscale modules into backbone CNNs significantly improves multimodal denoising performance through cross-scale and deep feature fusion strategies.
- **Residual and Cascade Approaches:** Removing the majority of noise first allows lightweight residual networks, attention mechanisms, or non-local modules to recover details.
- **Plug-and-Play Integration:** Encapsulating filtering layers as pluggable components allows seamless combination with convolutional, attention-based, or non-local blocks, supporting a variety of deep network architectures.

5.2 Transform Domain

Transform-domain denoising is no longer isolated from neural networks. Instead, it has been fully integrated into deep learning frameworks in an end-to-end fashion. In the wavelet domain, iterative thresholding unrolling transforms the classical wavelet shrinkage [169] process into a sequence of network layers, where each layer performs wavelet thresholding and residual correction. In the Fourier domain, the network front-end applies a fixed or differentiable FFT [170], while the core modules learn spectral masks or Wiener gains for each frequency component. In the DCT domain, the transform is interpreted as a shallow differentiable CNN [171]. The frequency coefficients undergo two convolutional layers for residual estimation before inverse transformation.

In the wedgelet domain, traditional predefined segmentation directions are replaced by a lightweight sub-network [172] that predicts the optimal partition angle or position for each block. In the curvelet domain, the sub-band thresholding is combined with fast NLM [173], and a lightweight network module is used to learn the optimal threshold parameters. In the bandlet domain, the noisy image is first transformed [174] using a bandlet transform and then fed, along with the residual input, into a compact CNN for learned compensation. In the contourlet domain, the non-subsampled contourlet transform analysis filters are used as non-trainable front-end modules [175] and these sub-bands are then passed into Transformer or non-local attention networks to learn content-adaptive, context-aware weighting.

Deep learning based classic frequency-domain filters are also highly effective for infrared image denoising. For instance, Saragadam et al. [176] introduce a framework that factors out thermal sensor non-uniformities via a differentiable FFT front end, then learns a deep CNN regularizer on the reconstructed spectrum before performing the inverse FFT. Frequency-domain filtering inherently analyzes the spectral distribution of the entire image, making it well suited to suppress periodic stripe noise and other global disturbances. By placing fixed or learnable FFT-DCT layers at the network input, these noise components can be aggressively attenuated in the spectral domain, leaving only fine details for subsequent sub-networks to recover.

Similarly, SMEDS Hu et al. [177] uses learnable down- and up-sampling filters within a symmetric multi-scale encoder-decoder architecture that function as frequency-selective kernels. Parallel multi-scale pyramids and non-local attention modules can be attached to different frequency bands or subbands; an attention-based fusion then re-weights each spectral component to balance global consistency with preservation of local thermal textures, markedly improving the structural coherence of the denoised output. Moreover, [178] propose a “frequency-domain guided” stripe removal

method in which an FFT-based low-rank plus sparse decomposition isolates stripe artifacts, followed by a compact CNN to compensate residual errors, yielding robust destriping performance on real infrared imagery. By turning classical frequency-domain filters, such as Wiener gains or soft thresholding, into learnable parameters, the network can automatically adapt its spectral gains to various thermal noise distributions during end-to-end training, thus maximizing reconstruction fidelity.

In summary, classical frequency-domain filters, when integrated into deep learning frameworks, significantly enhance performance in denoising tasks. Combination techniques include: 1) *Fixed Transform Frontend with Residual or Attention-based Backend*: Classical analysis-synthesis transforms are inserted at the input and output stages of CNNs or Transformers. The DCT coefficients of the sub-band or block are fed into lightweight residual networks or self-attention modules for fine-grained detail refinement. 2) *Learnable and Differentiable Transform Layers*: On the one hand, wavelet, framelet, or DCT filters are designed as trainable layers, ensuring frame tightness via spectral normalization or norm-based regularization. On the other hand, high-frequency gain modules are implemented as differentiable layers, learning signal-to-noise ratios or gains at each frequency component. 3) *Unrolling of Transform-domain Iterative Models*: Thresholding steps such as ISTA, Bregman iterations, and Plug-and-Play priors are reformulated as network layers. The iterative process is unrolled into a finite number of stages, each comprising sub-band thresholding or frequency-domain filtering, with learnable thresholds and step sizes. 4) *Cross-scale and Non-local Fusion*: Multi-scale pyramids and non-local self-attention modules are connected in parallel to the transform-domain outputs. Attention weights are used to fuse coefficients across scales, balancing global consistency with local texture preservation.

5.3 Signal Decomposition

Recently, dictionary learning-based denoising has become popular. Integration and innovation proceed mainly as follows: Convolutional dictionary learning treats dictionary atoms as learnable convolutional kernels [179]. During training, the network jointly optimizes these kernels along with the residual modules in the backend, enabling automatic adaptation of the dictionary to the underlying data distribution. Unrolled neural networks [180] reformulate classical iterative sparse coding; each iteration of sparse inference and dictionary update is learnable. Trainable thresholds, step sizes, and update rules allow adaptive optimization during end-to-end training. Integrating dictionary learning as a plug-gable module within the iterative optimization frameworks of the ADMM [181] invokes a compact dictionary-based

sparse coding network in each iteration, providing data-driven regularization while maintaining the flexibility of model-based approaches. The dictionary integrated with Transformer serves as a global or local sparse prior [182]. These priors improve feature encoding and self-attention mechanisms by injecting structural sparsity into the attention flow or token representations.

Dictionary learning-based filters embedded within deep learning frameworks show particular suitability for infrared image denoising. In thermal imaging, noise characteristics, such as stripes, spots, or speckles, vary greatly between different application scenarios. By performing online dictionary updates during network training on thermal datasets, the model can automatically learn a set of convolutional atoms that optimally suppress these scene-specific noise patterns without manually tuning thresholds or specifying noise models. Sparse coding inherently decomposes the input into a small number of salient atoms and redundant noise components. By unrolling the traditional encode-threshold-reconstruct pipeline into a sequence of learnable layers, the network simultaneously restores fine thermal textures while discarding noise artifacts. For instance, Li et al. [183] decomposes infrared images into sparse dictionary subbands and applies a lightweight neural network to the concatenated sparse codes for joint denoising and fusion.

Similarly, the adaptation of K-SVD learns overcomplete dictionaries tailored to stripe and speckle noise in thermal maritime imagery [184]. Other hybrid strategies include combining spatial-domain and frequency-domain dictionaries. An initial frequency-domain dictionary layer removes global periodic noise, followed by a spatial-domain dictionary layer that eliminates local speckle; a downstream network jointly compensates both residuals to address noise at all scales. Moreover, to handle diverse noise types in a single model, training multiple dictionaries in parallel allows dynamically fusing their output via attention mechanisms. Such dictionary ensemble fusion architecture delivers robust denoising performance across a wide range of infrared imaging scenarios.

5.4 Limitations

This review on denoising methods identifies key limitations, especially within the research domain. Denoising methods are interdisciplinary and span areas such as signal processing, machine learning, and statistics. This presents challenges in cross-disciplinary applications, as some summarized methods are tailored to specific data or noise models and may not be effectively generalized to other scenarios.

Furthermore, the generalization capability of these methods poses limitations. Although some concise denoising methods may excel in particular datasets, their effectiveness

can diminish when used on others. This suggests that customized adjustments and optimizations are often necessary for diverse datasets to achieve optimal denoising results.

Finally, the evaluation criteria used in the study present a challenge. Relying solely on metrics such as PSNR and SSIM to evaluate denoising methods may not provide a fully objective assessment of their performance across different approaches.

6 Conclusion

The denoising of RGB-infrared images has attracted considerable attention and has shown significant progress in recent years. The paper reviews comprehensively the existing literature on denoising methods. Specifically, the review classifies these approaches into two main categories: domain-based and signal decomposition-based. The review introduces and summarizes each category, highlighting core ideas and representative techniques. In addition, experiments demonstrated the performance of these methods using noisy thermal and RGB images from publicly available benchmark datasets. By providing a well-organized overview of current RGB-infrared denoising methods, the review serves as a valuable resource for researchers interested in exploring this field and provides a solid foundation for further investigation. Furthermore, denoising technology development contributes to vision tasks such as object detection and semantic segmentation. Summarizing the performance of classical denoising techniques in infrared images can provide guidance and reference for research and practice in infrared image denoising, considering broad infrared scenarios such as industrial production, safety protection, transportation, and other practical applications.

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Declarations

Conflicts of Interest The authors have no competing interests to declare that are relevant to the content of this article.

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